Big Data Fundamentals: Storing Big Data

In today's data-driven world, data engineers must efficiently handle big data. With organisations leveraging data for strategic decisions, proficiency in managing and analyzing big data is indispensable, driving the need for **scalable, efficient solutions** that transform data into **actionable insights**.

In this lesson, we'll explore the essentials of big data, including its defining **characteristics**, the challenges of **managing vast datasets**, and the solutions modern **data storage technologies** offer. We'll learn about the distinction between **data lakes and warehouses**, and how to design **big data strategies** that deliver significant business value. We'll build a solid foundation in big data principles and practices and an understanding of **data persistence in containerisation**, enabling us to navigate and implement effective data management strategies in various industries.



What You'll Learn:

Understanding Data Lakes and Warehouses

Explore the concepts of data lakes and warehouses, including their components, design considerations, and best practices.

Block and Object Storage Essentials

Gain knowledge of block and object storage and their relevance beyond traditional database systems.

Key Features of Data Warehouses in Business Intelligence

Analyze the features, benefits, and use cases of key data warehouses, understanding their role in business intelligence.

Data Persistence in Containerised Environments

Understand how data persistence is possible in a containerised environment.

Ready to Dive In?

Reflect:
Think about an app or service you use regularly — a streaming platform, a banking app, a social feed. Where do you think its data lives? Would a data lake, a data warehouse, or object storage be the right fit, and what about that data would drive that choice?